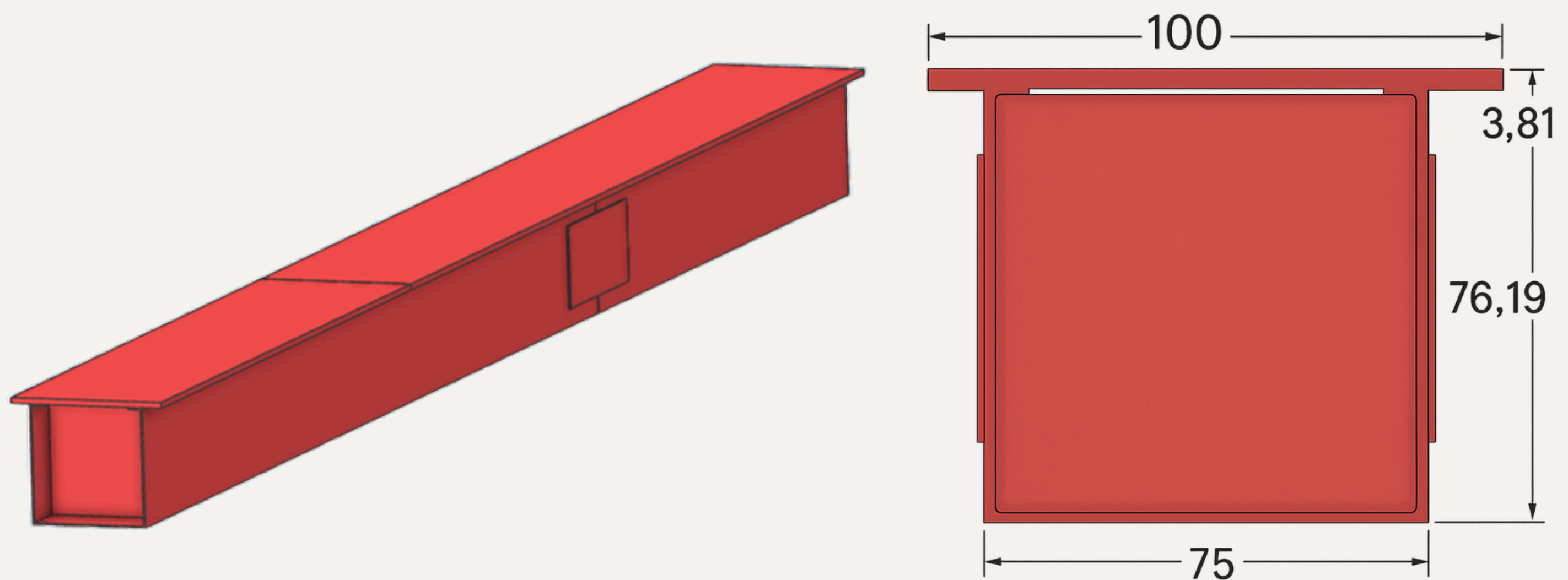


H O P E S A N D P R A Y E R S

MATBOARD AND CONCRETE BRIDGE



Hopes and Prayers is a Matboard and concrete bridge 1265mm in length. This bridge has a T-shaped cross sectional area to support a total of 400N. It also contains a splice connection at a 1 : 1.80 ratio to achieve lightweight purposes. Furthermore, this bridge contains only simple geometric shapes to maximize the construction quality. This bridge is analyzed via our team-developed MatLab Code.¹

What is the Matboard Bridge Competition?

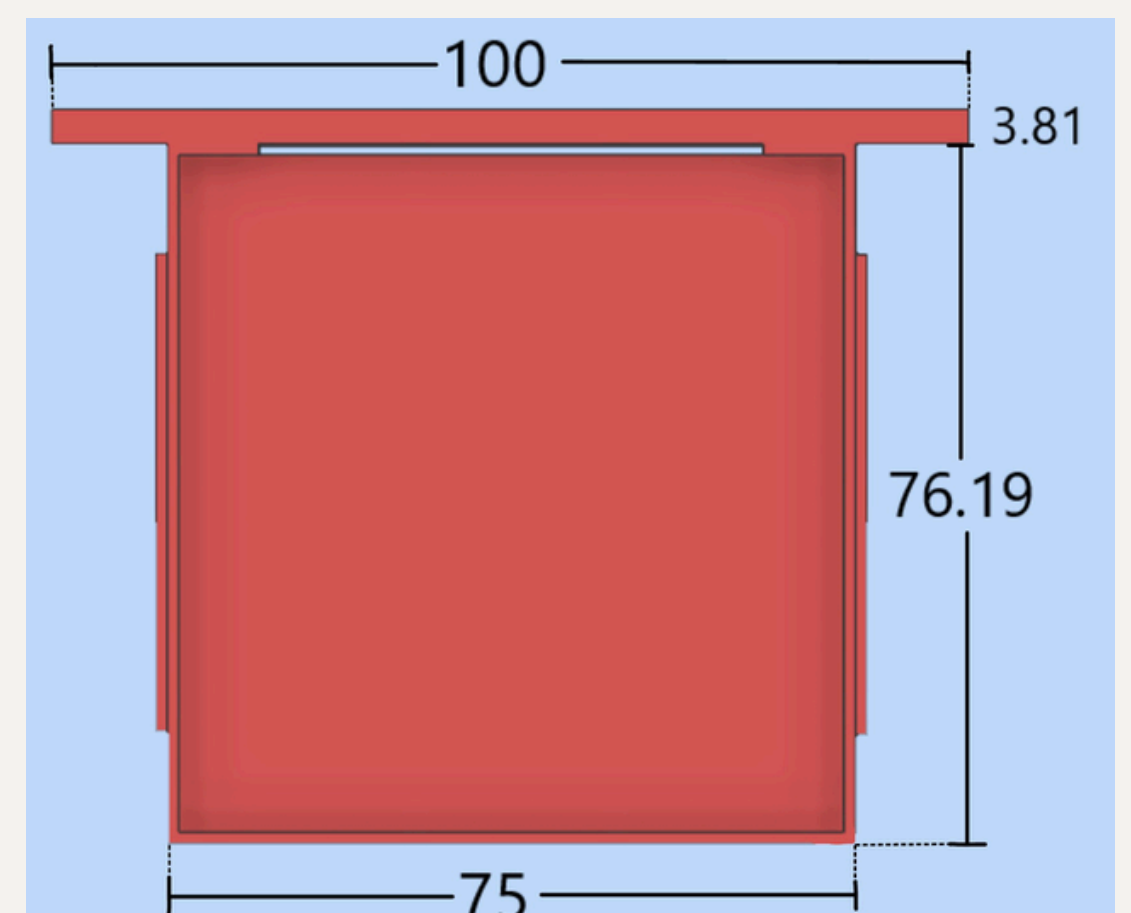
CIV102 is one of the most iconic and difficult courses of engineering science, the highlight of this course is the Matboard Bridge Competition.

In 2024 Fall, students construct bridges using only a 32" x 40" matboard and concrete glue. Bridges then are competed and evaluated based on strength to weight ratio.

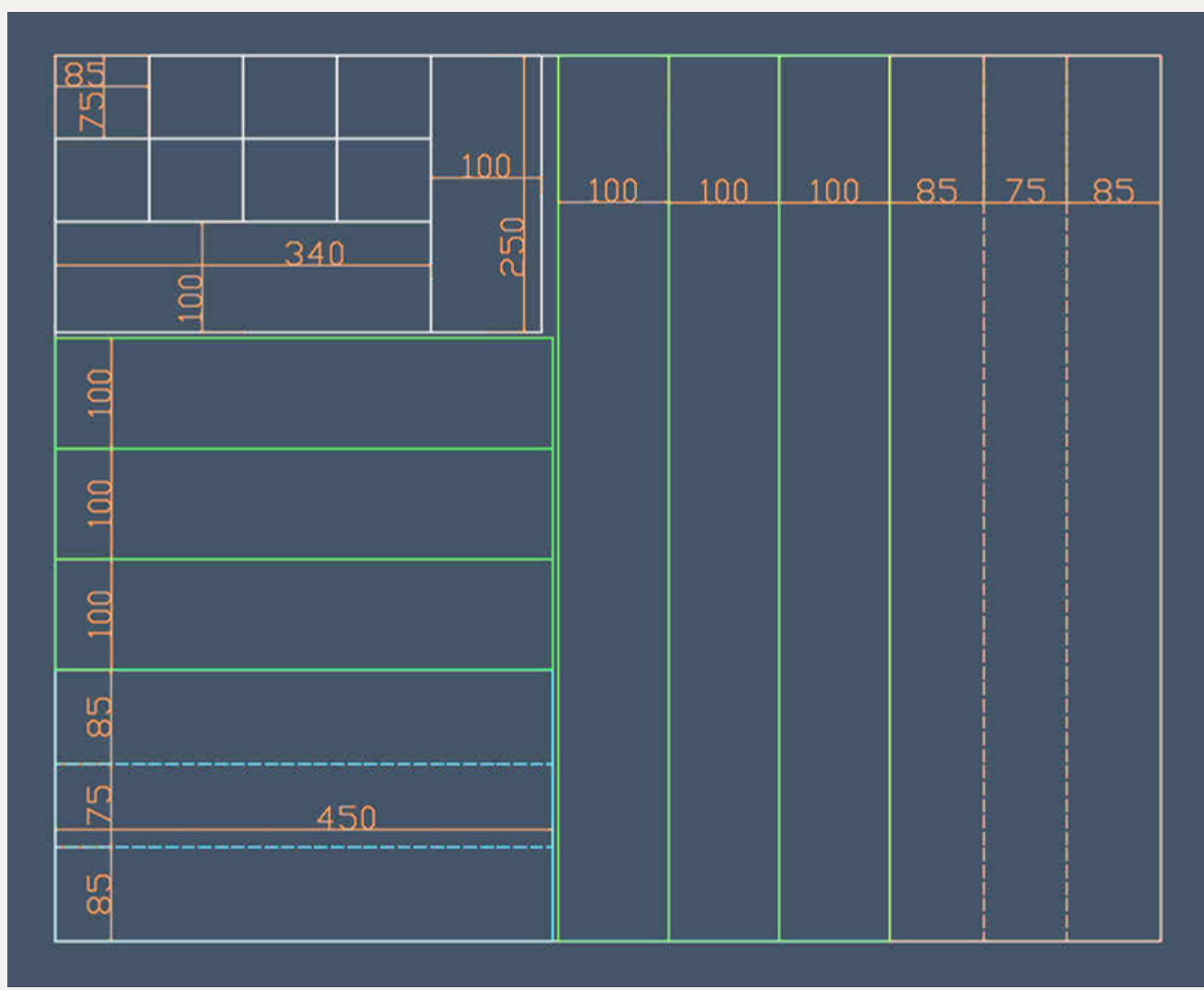
KEY DESIGN FEATURES

01 T-Shape Cross-Sectional Area

Through concepts such as Thin Plate buckling, beam theory and Moment of Inertia, it is determined that this cross-sectional area optimizes supportive weight the best compared to other designs.



¹: Special Thanks to Paul Choi for writing the MatLab Code



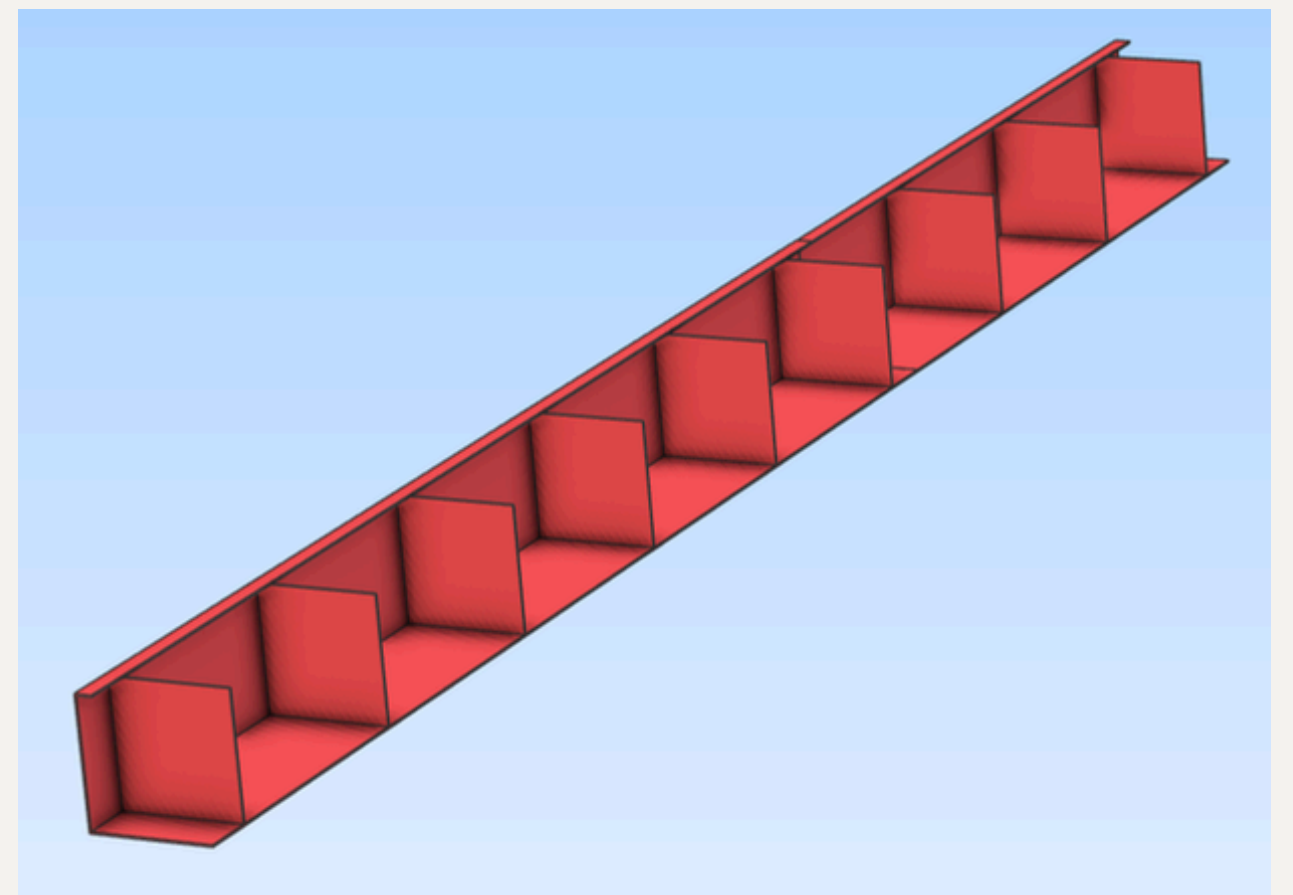
Splice Connection 02

Using AutoCAD, the team determined that a splice connection of 1: 1.8 ratio allows us to save the maximum amount of material. This only uses 31.3" x 39.1" instead of the whole Matboard.

03 Simple Geometric Shapes

Hopes and Prayers contains a structure formed by simply rectangles.

This is done to simplify the cutting process and maximize the build quality.



DESIGN VALUES

Quality

- The bridge has almost perfect quality, with Prof. Evan Bentz giving it a score of 9/10.

Strength

- While more than 50% of bridges fail at 400N, our bridge failed at 430N.
- 400N is approximately the weight of a 10 year old.

Lightweight

- Our bridge contained ~ 4.1% leftover materials.
- Matboard has a low density of $\sim 1\text{g/cm}^3$.

POSSIBLE IMPROVEMENTS

More Iterations

Load Case 2
 FOS for tension 4.3065
 FOS for compression: 2.0613
 FOS for flexural buckling: 12.153
 Case 1 buckling
 FOS for shear: 2.3029
 FOS for glue: 10.0772
 FOS for shear buckling: 3.1387

Currently, our bridge has the lowest factor of safety (FOS) for compression.

In the future, the cross-sectional area can be more optimized to raise the FOS.

Stronger Splicing



On the actual testing day, our bridge failed due to shear stress on the splicing.

In the future, enhancing the splice connection ensures a greater strength supported by the bridge.